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A bio-inspired framework for apple leaf disease detection: Integrating lesion localization, ant colony optimization, and machine learning

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ABSTRACT

Apple trees, among the most widely cultivated and economically important orchard species, are highly susceptible to foliar diseases such as Black Spot, Black Rot, and Cedar Rust. Due to the visual similarity of symptoms, accurately distinguishing among these diseases poses a major challenge. Conventional diagnostic approaches, such as expert visual assessments and laboratory analyses, are often time-consuming, costly, and limited to post-symptomatic stages. To address the growing need for rapid, accurate, and scalable solutions in precision disease detection and management, this study presents a novel framework integrating image processing, artificial intelligence (AI), and ant colony optimization (ACO) for automated disease classification in apple leaves. The proposed method comprises five key steps: (1) background removal from leaf images, (2) diseased area detection, (3) extraction of texture, color, and shape features, (4) feature selection using ACO to identify the most informative attributes, and (5) disease classification using a support vector machine (SVM) classifier. Experimental results demonstrate that preprocessing steps, particularly background removal and lesion localization, significantly enhance classification accuracy. The system achieved class-wise accuracies of 95.12 % (Black Spot), 90.91 % (Black Rot), 94.87 % (Cedar Rust), and 88.89 % (Healthy), with an overall classification accuracy of 92.50 %. Among the features used, texture contributed most significantly to performance, followed by color and shape. These findings highlight the effectiveness of combining diverse image features with bio-inspired optimization techniques for plant disease detection and offer a promising direction for future research and deployment in intelligent agricultural monitoring systems.

1. Introduction

Apple cultivation is a cornerstone of the global horticultural industry, contributing significantly to agricultural economies worldwide. However, apple production is frequently threatened by a range of foliar diseases, particularly Black Spot, Black Rot, and Cedar Apple Rust, which can severely impact yield and fruit quality. A key challenge in managing these diseases lies in their overlapping visual symptoms, which complicates accurate diagnosis and timely intervention [1]. Apple Black Spot, for example, manifests as olive green to brown lesions on the leaves. As the infection progresses, the affected foliage often turns yellow and falls prematurely, especially under high-humidity conditions where the fungus thrives. Similarly, Black Rot, which can present alongside Frog Eye Leaf Spot, produces small brown spots or necrotic lesions, often accompanied by tiny perforations in the leaf surface. These similarities make it difficult to distinguish between the two diseases through visual inspection alone [2]. Cedar Apple Rust further complicates the diagnostic landscape by inducing yellow-orange lesions not only on the leaves but also on branches and fruit, creating symptomatic overlaps with other common fungal diseases [3]. The high degree of visual similarity among these diseases, along with their co-occurrence in many orchards, often leads to misdiagnosis [4]. If not correctly identified and treated at an early stage, these diseases can lead to substantial crop losses and diminished fruit marketability. As such, early, accurate, and automated disease diagnosis is critical for enabling timely interventions and reducing economic losses in apple production [5–7].

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Traditional methods for plant disease assessment, typically performed by experts, are often labor-intensive, time-consuming, and costly, with diagnostic accuracy largely dependent on the individual's experience and subjective judgment [8,9]. These expert-based evaluations are susceptible to human error, especially under conditions of fatigue, inconsistent observation, or when disease symptoms are subtle or visually similar. In the context of modern agriculture, where efficiency, scalability, and precision are critical, such manual approaches are increasingly inadequate [10–12]. Although laboratory-based diagnostic techniques offer higher accuracy, they present their own set of limitations. Common methods such as pathogen isolation, serological assays, and polymerase chain reaction (PCR) testing require sophisticated equipment, trained personnel, and extended processing times. Furthermore, these techniques are typically applicable only after visible symptoms have emerged, thereby limiting their utility for early-stage disease detection; a critical factor for effective management and control [13,14].

Given the growing demand for fast and reliable plant disease diagnosis, the development of innovative and automated approaches, including advanced image processing and artificial intelligence (AI) techniques, has become essential. These modern methods can overcome the limitations of traditional visual and laboratory approaches, offering faster, more accurate, and scalable solutions for disease detection in agriculture [15,16].

Within this evolving landscape, metaheuristic optimization algorithms and evolutionary computing techniques have emerged as powerful tools for solving complex feature selection problems, which are critical to improving classification performance in AI-driven disease detection systems [17,18]. Numerous studies have demonstrated the effectiveness of these approaches. For example, Kumar et al. (2020) extracted 38 color, texture, and shape features from apple leaf images to classify diseases such as powdery mildew, mosaic, and rust. By applying a genetic algorithm combined with correlation-based feature selection, they reduced dimensionality and improved accuracy to 90 %. In another study, Cristin et al [19] proposed an image processing framework where noise removal was followed by fuzzy clustering to segment regions of interest. Texture features were extracted using histograms of oriented gradients (HOG) and entropy measures, and classification was performed using a hybrid model that combined deep learning with the cuckoo search algorithm, achieving 87 % accuracy. Similarly, Javidan et al [20] developed a novel algorithm to detect four types of apple leaf diseases, reporting a classification accuracy of 91 %. These findings highlight the potential of integrating optimization algorithms with AI and image-based feature analysis to advance the precision and efficiency of automated plant disease detection systems.

Building on recent technological advancements, AI and image processing have become pivotal in developing automated systems for plant disease detection. Through image processing, critical visual features such as color, texture, and shape can be extracted from leaf images, enabling detailed analysis of disease symptoms [21]. Meanwhile, AI techniques, particularly machine learning and deep learning, have shown great promise in accurately classifying plant diseases based on these features [22]. However, one of the key challenges in these systems is the high dimensionality of the extracted features, which can degrade model performance, increase computational complexity, and introduce noise. To overcome this limitation, feature selection plays a crucial role by identifying the most informative features and eliminating those that are redundant or irrelevant. Among various feature selection strategies, ant colony optimization (ACO) has gained considerable attention due to its robustness and efficiency in navigating complex search spaces. Inspired by the collective foraging behavior of ants, ACO is a bio-inspired metaheuristic algorithm capable of finding near-optimal solutions to combinatorial optimization problems, including those in high-dimensional feature spaces [22].

In this study, a novel approach was developed that integrates ACO-based feature selection with AI-driven classifiers and image processing techniques to improve the detection and classification of visually similar apple leaf diseases: Black Spot, Black Rot, and Cedar Apple Rust. The proposed method involves five key stages: background removal, diseased area detection, multi-feature extraction (color, texture, and shape), ACO-based feature selection, and final classification using a support vector machine (SVM) classifier. The results demonstrate significant improvements in classification accuracy and computational efficiency, particularly when compared to models without feature optimization. The framework not only addresses existing limitations in disease detection but also contributes to the broader goals of precision agriculture and sustainable farming by enabling scalable, automated, and reliable disease diagnosis in orchard systems.

2. Materials and methods

Developing accurate image-based classifiers for plant disease diagnosis requires access to a robust and verified dataset comprising images of both healthy and diseased leaves. Historically, such datasets were either unavailable or difficult to access in the public domain. To address this challenge, this study leveraged the PlantVillage dataset [23], a widely used and freely accessible image repository designed to support the development of AI-based plant disease diagnostics. The dataset includes a diverse collection of high-resolution leaf images captured under controlled conditions and categorized by plant species and disease type.

For this study, a subset of the PlantVillage dataset was used, consisting of apple leaf images grouped into four categories: Healthy, Black Spot, Black Rot, and Cedar Apple Rust. These three disease classes were selected due to their visual similarity, which presents a realistic and challenging classification task for automated diagnosis systems. The proposed classification framework consists of five main stages, as shown in Fig. 1: (1) Preprocessing: backgrounds are removed from the original images to isolate leaf regions and reduce noise. (2) Disease Area Detection: Infected regions are identified to focus analysis on symptomatic areas. (3) Feature Extraction: A comprehensive set of color, texture, and shape features is extracted from the diseased regions. (4) Feature Selection: The ACO algorithm is employed to select the most relevant features and reduce dimensionality. (5) Classification: An SVM classifier is used to assign each image to one of the four defined classes. Each of these steps is described in detail in the following sections.

2.1. *Preprocessing: background removal and diseased region isolation*

The preprocessing stage is essential for isolating the leaf from its background, thereby enhancing the accuracy of subsequent disease detection and feature extraction. This process consists of a sequence of image processing steps designed to generate a clean leaf mask and facilitate the segmentation of diseased and healthy regions (Fig. 2a–h):

(a) Original Image: The process begins with the raw RGB image, which includes both the leaf and background.
(b) Green Channel Extraction: The green color band is extracted to emphasize the leaf region and suppress background noise, leveraging the leaf's natural pigmentation.
(c) Edge Detection: An edge detection algorithm is applied to detect the contours of the leaf.
(d) Morphological Dilation: Dilation operations are used to bridge any gaps in the detected edges, ensuring a closed boundary around the leaf.
(e) Hole Filling: Internal gaps within the leaf contour are filled to generate a solid binary mask.
(f) Background Removal: The binary mask is applied to the original image, effectively removing the background and retaining only the leaf.
(g) Disease Region Segmentation: Image segmentation techniques are applied to isolate symptomatic (diseased) areas on the leaf surface.

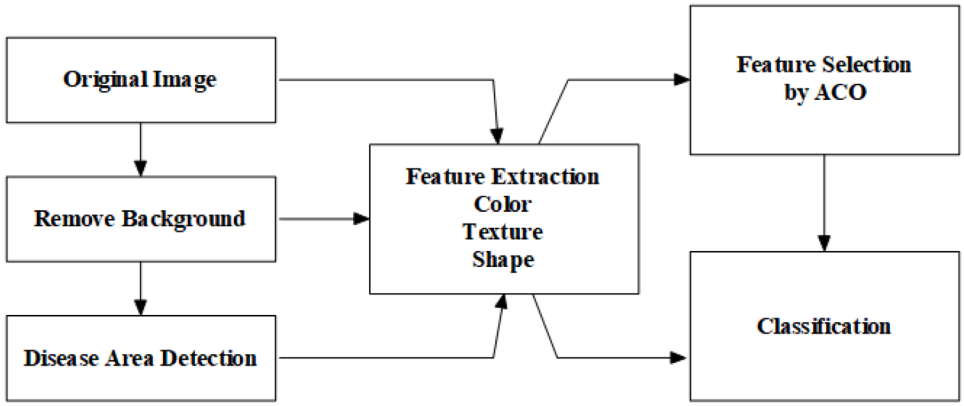


Fig. 1. Overview of the proposed methodology for automated classification of apple leaf diseases using image processing, ACO-based feature selection, and SVM classification.

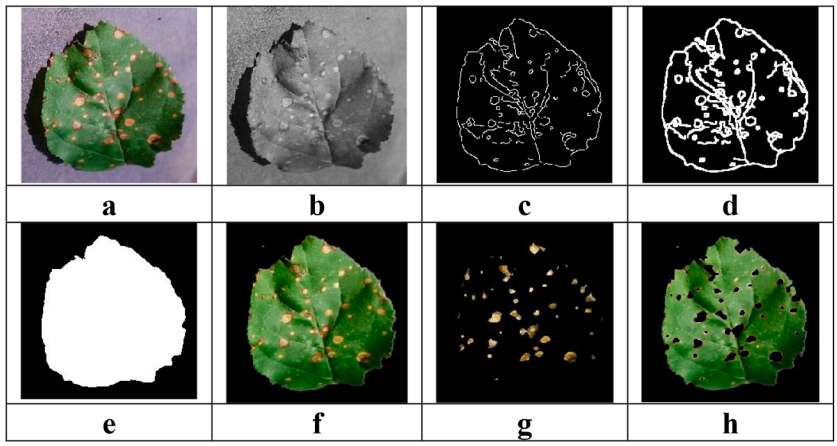


Fig. 2. Workflow of background removal and disease segmentation from apple leaf images: (a) Original image, (b) Green channel extraction, (c) Edge detection, (d) Morphological dilation, (e) Hole filling, (f) Background removed, (g) Diseased region segmented, (h) Healthy area identified.

(h) Healthy Area Isolation: Remaining regions of the leaf not identified as diseased are labeled as healthy, enabling comparative analysis.

This preprocessing pipeline ensures consistent background removal and effective separation of diseased and healthy tissue, providing a reliable foundation for feature extraction and classification [24].

2.2. Disease region segmentation on apple leaves

Image segmentation plays a critical role in isolating diseased regions from healthy tissue. Techniques range from simple thresholding to more advanced frequency domain methods. Among these, K-means clustering and Otsu's thresholding are commonly used for their efficiency and effectiveness [25,26]. In this study, the K-means clustering algorithm was employed to automatically segment diseased areas based on color differences. Healthy regions, primarily green, were masked out by excluding pixels where the red channel intensity was lower than both the blue and green channels. This approach effectively isolated symptomatic regions for further analysis.

This segmentation process isolated only the diseased regions in the final images. The K-means clustering algorithm, a widely used unsupervised learning method, partitions pixels into K clusters by minimizing the sum of squared distances between each pixel and its assigned cluster center. The algorithm iteratively performs four steps: (1) initializing cluster centers, (2) assigning pixels to the nearest cluster, (3) updating cluster centers based on the mean of assigned pixels, and (4) repeating the assignment and update steps until convergence. Traditionally, selecting the number of clusters (K) and manually defining the region of interest (ROI) introduced inefficiencies and inconsistencies. In prior approaches, each image required manual ROI selection to isolate diseased areas, increasing labor and the potential for human error. To overcome these limitations, this study automated ROI selection through color-based K-means clustering. Diseased areas were identified by excluding pixels where the red channel intensity was lower than that of the green and blue channels, effectively masking healthy regions. This automated approach significantly reduced processing time and improved the consistency and accuracy of disease area detection. Fig. 3 presents the sequential stages of the image processing pipeline used in this study.

2.3. Feature extraction

The feature extraction process involved analyzing color, shape, and texture properties of images to extract meaningful data for disease classification. Color features such as mean, maximum, standard deviation, and median were calculated across the RGB, HSV, l*a*b, and YCbCr color spaces to capture variations in color intensity and distribution within the images [27]. These features provide a comprehensive representation of the color features in the diseased and healthy regions. For better visualization, the four color spaces are shown in Fig. 4.

For textural features, the gray-level co-occurrence matrix (GLCM) was employed to compute metrics such as correlation, contrast, homogeneity, standard deviation, energy, entropy, variance, smoothness, root

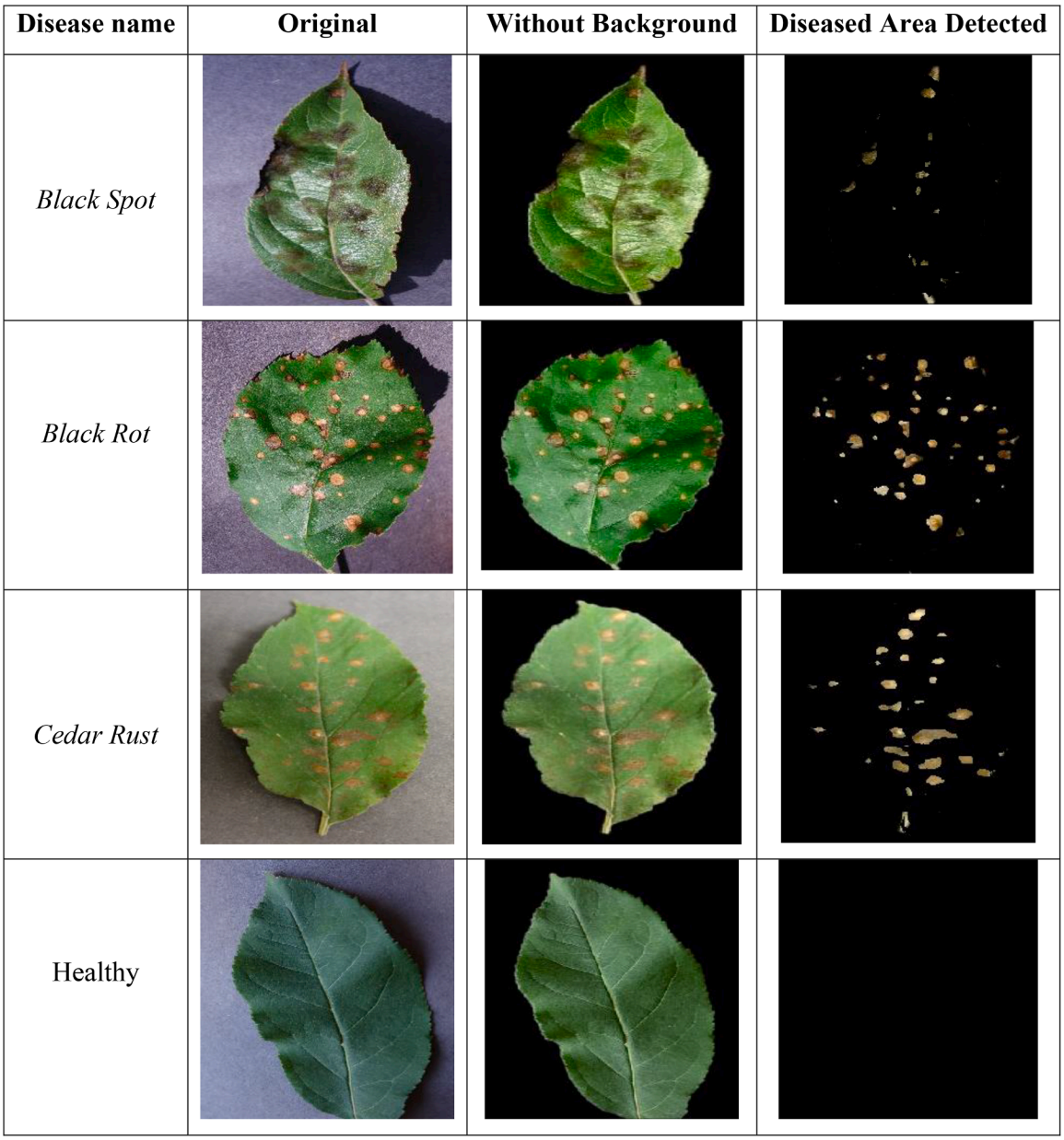


Fig. 3. Background removal and disease area detection.

mean square (RMS), kurtosis, skewness, mean, and the inverse difference moment (IDM). These features reflect the surface patterns and spatial relationships between pixel intensities, offering critical insights into the texture of diseased and healthy leaf areas. Each textural feature was calculated individually for every band in the aforementioned color spaces to ensure a detailed analysis [28].

In addition to color and texture, shape features were extracted to describe the geometry of the diseased spots. Metrics such as the major and minor axis lengths of the spots, perimeter, area, the number of detected objects, and the circularity index were calculated. These shape-based features help quantify the size, structure, and distribution of diseased regions, providing essential information for distinguishing between different types of diseases. By combining these color, texture, and shape features, the extracted data offers a robust foundation for accurate disease classification [13,29]

2.4. Feature selection by ant colony optimization (ACO) algorithm

Ant colony optimization (ACO) is a nature-inspired optimization algorithm based on the foraging behavior of ants. It is particularly effective for solving complex optimization problems, including feature selection in machine learning. In the context of feature selection, ACO is used to identify the most relevant features from a large set of candidate features. The goal is to select a subset of features that optimally improves the performance of a model, such as a classifier [30].

In ACO, artificial ants explore the solution space by selecting features based on pheromone trails, which represent the desirability of selecting certain features. Initially, ants choose features randomly, but over time, the selection process is influenced by the accumulated pheromone information. Features that contribute to better classification performance are reinforced with higher pheromone levels, making them more likely to be selected in subsequent iterations. Through this iterative process, the algorithm converges to an optimal or near-optimal feature subset [31]. The basic steps in ACO for feature selection are (Fig. 5):

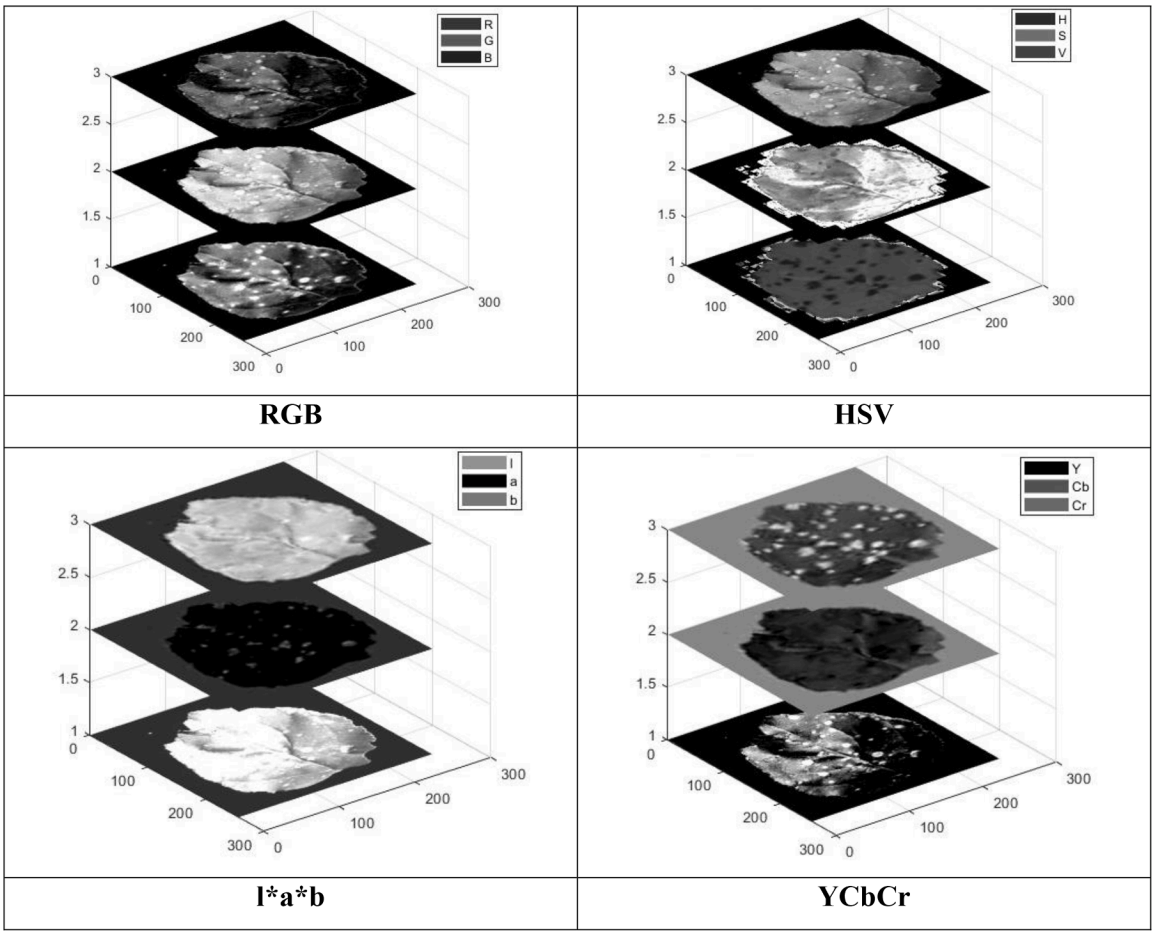


Fig. 4. Different slices of images in different color spaces.

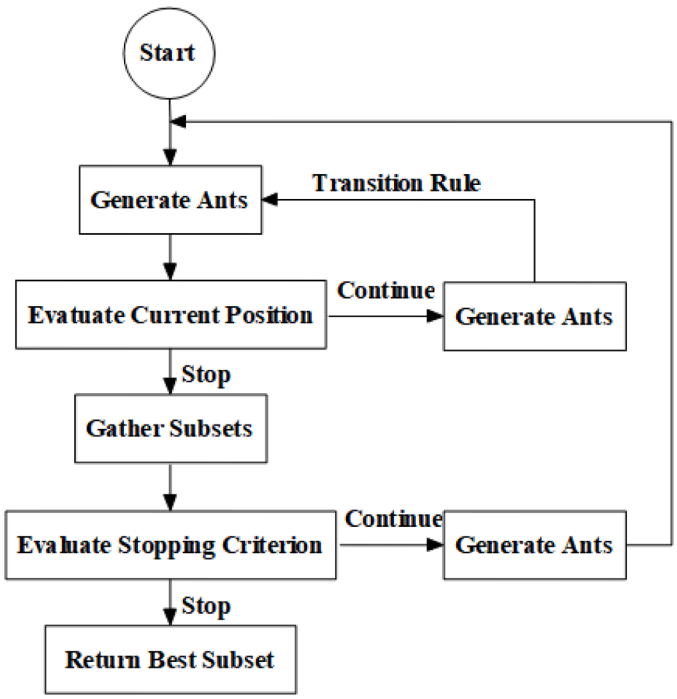


Fig. 5. ACO-based feature selection overview.

1. **Initialization**: Create a population of ants, each representing a possible solution (i.e., a subset of features). Initialize pheromone values on all features.
2. **Ant Movement**: Each ant chooses a set of features based on a probabilistic decision rule influenced by pheromone values and feature relevance.
3. **Solution Evaluation**: Evaluate the quality of the selected feature subset using a fitness function (e.g., classification accuracy, error rate, etc.).
4. **Pheromone Update**: Update pheromone values based on the quality of the solutions. Features that contribute to better performance are reinforced, while those that do not are eliminated.
5. **Iteration**: Repeat the process for a predefined number of iterations or until convergence is achieved.

The process continues until the optimal feature subset is found, which minimizes or maximizes the fitness function (depending on the problem).

The following variables can be defined for this optimization algorithm [32,33]:

- $F = \{f_1, f_2, \ldots, f_n\}$ as the set of all features.
- S_k, represents the feature subset selected by the k-th ant.
- $P(f_i)$ is the probability that feature f_i is selected by an ant, which is determined by the pheromone value on f_i, denoted by $\tau(fi)$, and a heuristic function $\eta(fi)$ that measures the relevance of the feature.
- $\tau(fi)$ is the pheromone level associated with feature fi.
- $\eta(fi)$ is a heuristic value based on the usefulness of the feature (such as information gain or correlation with the target variable).

The probability of selecting feature *fi* by the *k*-th ant is given by:

$$P(f_i) = \frac{[\tau(f_i)]^{\alpha}.[\eta(f_i)]^{\beta}}{\sum_{j=1}^{n}\left[\tau\left(f_j\right)\right]^{\alpha}.\left[\eta\left(f_j\right)\right]^{\beta}} \quad (1)$$

Where: α and β are parameters that control the influence of pheromone and heuristic information, respectively.

After the ants select their feature subsets, the quality of each subset is evaluated using a fitness function. Based on the performance, the pheromone values are updated using the following equation:

$$\tau(f_i) = (1-\rho).\tau(f_i) + \Delta\tau(f_i) \quad (2)$$

Where: ρ is the pheromone evaporation rate, $\Delta\tau(fi)$ is the pheromone deposit on feature *fi*, which is a function of the fitness value of the solution containing *fi*.

Finally, the algorithm repeats the process, refining the feature selection through iterations until convergence is achieved. The selected feature subset at the end of the process is the one with the highest accumulated pheromone value, indicating it is the most optimal for the task at hand [34].

2.5. Classification methods

Classification is a fundamental task in data mining and machine learning that involves assigning class labels to input samples. In this study, a linear SVM was employed to classify images of apple leaves as either healthy or diseased. Linear SVMs offer significantly faster training times compared to non-linear alternatives, especially when the data is linearly separable or contains many features [13]. Given these characteristics in our dataset, a linear kernel was chosen for both detection and classification tasks. A confusion matrix was used to evaluate classification performance. To ensure balanced data, 100 images were selected for each class, and the dataset was split into training and testing sets using an 80:20 ratio.

2.6. Performance evaluation criteria

The performance of the models was evaluated using several criteria: accuracy, precision, sensitivity, specificity, and F-measure. These metrics provide a comprehensive assessment of the model's effectiveness in classifying the data [35].

- **Accuracy** measures the overall correctness of the model, calculating the proportion of correct predictions (both true positives and true negatives) out of all predictions.
- **Precision** evaluates the proportion of true positives out of all predicted positives, indicating how many of the predicted positive cases were actually correct.
- **Sensitivity** (also known as recall) measures the proportion of true positives out of all actual positives, reflecting the model's ability to identify all relevant instances.
- **Specificity** calculates the proportion of true negatives out of all actual negatives, highlighting the model's ability to correctly identify non-relevant instances.
- **F-measure** is the harmonic mean of precision and sensitivity, providing a single metric that balances both concerns.

These criteria were used to comprehensively assess the performance of the models and determine their effectiveness in detecting and classifying plant diseases.

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)} \quad (3)$$

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

$$Sensitivity = \frac{TP}{TP+FN} \quad (5)$$

$$Specificity = \frac{TN}{TN+FP} \quad (6)$$

$$F-Measure = \frac{2\times(Precision\times Sensitivity)}{(Precision+Sensitivity)} \quad (7)$$

Where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively.

3. Results and discussion

3.1. Feature selection by the ACO algorithm

Selecting the significant features is a crucial component of machine learning and statistical pattern recognition, as it significantly affects the precision, efficiency, and overall functionality of the system. This step becomes even more important in fields like data mining and intelligent modeling, where irrelevant or weakly associated features can undermine the system's performance. Acting as a foundational stage in the development of intelligent learning models, feature selection helps to manage high-dimensional datasets by narrowing the feature space, which in turn reduces the computational burden during training [36]. Eliminating redundant and non-contributive features not only simplifies data processing but also helps reduce dimensionality [37].

In this research, 68 features were extracted from RGB images. These consisted of 48 features related to color, 14 capturing texture, and 6 describing shape. To identify the most relevant and impactful features for distinguishing between healthy and diseased leaves, the ACO algorithm was used. Six important features were ultimately selected from each feature group: color, texture, and shape. Table 1 presents the top-performing features identified by the ACO algorithm within each category.

The analysis of Table 1 highlights the discriminative power of different feature groups, color, texture, and shape, in the classification of apple tree leaf diseases, including Black Spot, Black Rot, Cedar Rust, and healthy leaves. Among these, texture features demonstrated the highest classification accuracy (80.15 %), indicating their strong capacity to capture disease-specific surface variations. Features such as entropy, energy, and skewness ranked highest within the texture group. Entropy, for example, quantifies the randomness or complexity within the image and is highly sensitive to the irregular patterns caused by fungal infections. Energy, on the other hand, reflects textural uniformity and can effectively distinguish between smooth healthy leaf surfaces and the disrupted textures found in diseased regions.

Color features followed closely, achieving an accuracy of 75.62 %, and are particularly informative due to the visual color changes that diseases impose on leaf tissues. The top-ranked feature, standard deviation in the Y channel of the YCbCr color space, indicates how brightness varies within the leaf, capturing the subtle tonal shifts associated with infections like Cedar Rust or Black Rot. Median values in YCbCr and HSV spaces, as well as maximum intensity in HSV band V, are sensitive to chlorosis or necrotic spots, which typically appear as brightness or hue variations. The presence of Median a from the l*a*b space further supports the ability of chromatic variation to reflect disease stress, particularly reddish or purplish discoloration common in certain infections.

Shape features, while biologically relevant, showed the lowest classification accuracy at 63 %, indicating that morphological changes, although indicative, are less consistent or discriminative across different apple leaf diseases. Nevertheless, features such as the number of segmented objects and lesion area may still contribute to identifying

Table 1
Top 6 features in the classification of apple tree leaf diseases.

Feature rank	1	2	3	4	5	6	Accuracy %
Color feature	Standard Deviation YCbCr band Y	Median YCbCr band Y	Max HSV band V	Median HSV band V	Median l*a*b band a	Median RGB band B	75.62
Texture feature	Entropy	Energy	Skewness	Kurtosis	Contrast	Homogeneity	80.15
Shape feature	Number of objects	Area	Perimeter	Major axis length	Minor axis length	Circularity Index	63

severely infected leaves, where significant structural deformation is present due to extensive tissue damage. For example, an increase in the number of segmented regions could reflect multiple infection sites as seen in Black Spot disease, whereas major and minor axis lengths relate to general leaf elongation or contraction, which may occur in advanced stages of disease. Texture and color features appear to be the most effective for distinguishing between healthy and diseased leaves, due to their sensitivity to pigmentation and surface alterations caused by pathogens. Shape features, while useful in specific cases, may need to be complemented by other modalities or used in combination with texture and color information to improve overall classification performance

Recent studies have demonstrated that manually extracting features, also known as feature engineering, plays a crucial role in enhancing the accuracy of plant disease classification systems [38]. Moreover, selecting high-ranking features contributes to reducing computational demands and speeds up the classification process [39,40]. Among various feature types, texture-based descriptors are particularly effective due to their sensitivity to subtle surface variations commonly associated with plant diseases. Color features serve as a complementary tool by capturing changes in leaf pigmentation that often signal disease onset. Shape-related features, while generally more informative in later stages of infection, provide essential geometric context. The integration of these three feature types forms a solid and efficient basis for reliable plant disease detection [41].

3.2. Apple leaf disease classification

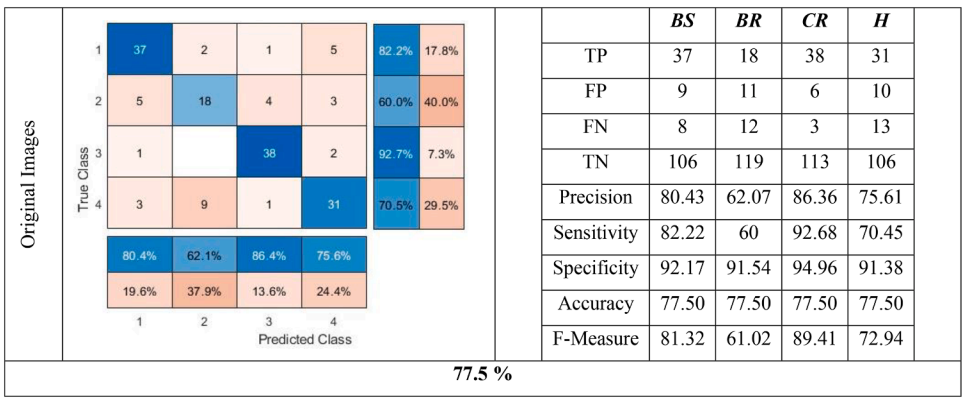
After extracting relevant color, texture, and shape features from the original images using the ACO algorithm, the selected features were combined and used to classify the apple leaf diseases with a linear SVM classifier. The classification results are presented as confusion matrices in Figs. 6, 7, and 8 (left). Class labels were assigned as follows: Black Spot (BS) = 1, Black Rot (BR) = 2, Cedar Rust (CR) = 3, and Healthy (H) = 4. In addition to the confusion matrices, Figs. 6, 7, and 8 (right) also display key performance metrics, including the false negative rate (FN), false positive rate (FP), true positive rate (TP), and true negative rate (TN).

3.2.1. Classification of apple leaf diseases using original images

The classification of plant diseases from the original images has shown considerable promise but also highlighted areas for further improvement. Initially, when classifying images of apple leaves with disease, the model demonstrated a good performance across several metrics, such as precision, sensitivity, specificity, and accuracy. Precision, which reflects the accuracy of positive predictions, was notably high for Cedar Rust at 86.36 % and for Black Spot at 80.43 %, indicating the model's strong ability to correctly identify these diseases. However, Black Rot had a significantly lower precision of 62.07 %, suggesting that the model often misclassified this disease, likely due to its visual similarities with other diseases. The model performed reasonably well in classifying healthy leaves (H) with a precision of 75.61 %, though there was still room for improvement. Sensitivity, which measures how well the model detects true positive cases, was highest for Cedar Rust at 92.68 %, but dropped for Black Rot to 60 %, indicating the model missed many instances of this disease. Specificity, a measure of how well the model identifies non-diseased leaves, was high across all categories, with Cedar Rust achieving the best score of 94.96 %, highlighting the model's capacity to distinguish healthy leaves from this disease. While the model showed consistent accuracy at 77.50 % across all diseases, the F-measure, which balances precision and sensitivity, indicated that improvements were needed, especially for Black Rot.

3.2.2. Classification of apple leaf diseases using background-removed images

Removing the background from the images resulted in a notable improvement in classification performance across all disease categories. This preprocessing step effectively eliminated background noise, enabling the model to concentrate on disease-relevant regions of the leaf, thereby enhancing feature discrimination. Precision improved significantly, particularly for diseases with complex or subtle symptoms. For example, precision for Black Rot increased from 62.07 % to 85.71 %, and for Cedar Rust, it rose from 86.36 % to 93.02 %. These gains suggest that background elements previously contributed to misclassification, especially for visually ambiguous cases like Black Rot. Sensitivity also improved across the board. Black Rot sensitivity increased from 60 % to 78.95 %, and Black Spot from 82.22 % to 91.18 %, indicating a more



	BS	BR	CR	H
TP	37	18	38	31
FP	9	11	6	10
FN	8	12	3	13
TN	106	119	113	106
Precision	80.43	62.07	86.36	75.61
Sensitivity	82.22	60	92.68	70.45
Specificity	92.17	91.54	94.96	91.38
Accuracy	77.50	77.50	77.50	77.50
F-Measure	81.32	61.02	89.41	72.94

77.5 %

Fig. 6. Classification of apple leaf diseases using original images: Confusion matrix (left) and evaluation criteria (right).

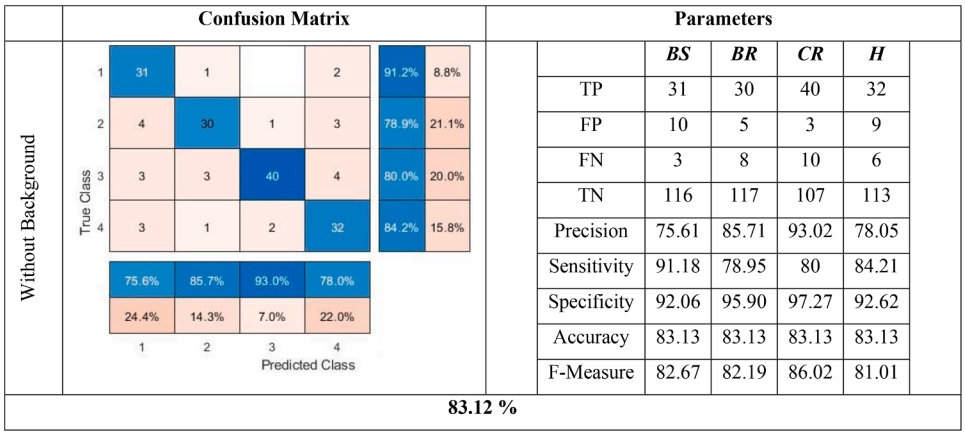


Fig. 7. Classification of apple leaf diseases using background-removed images: Confusion matrix (left) and evaluation criteria (right).

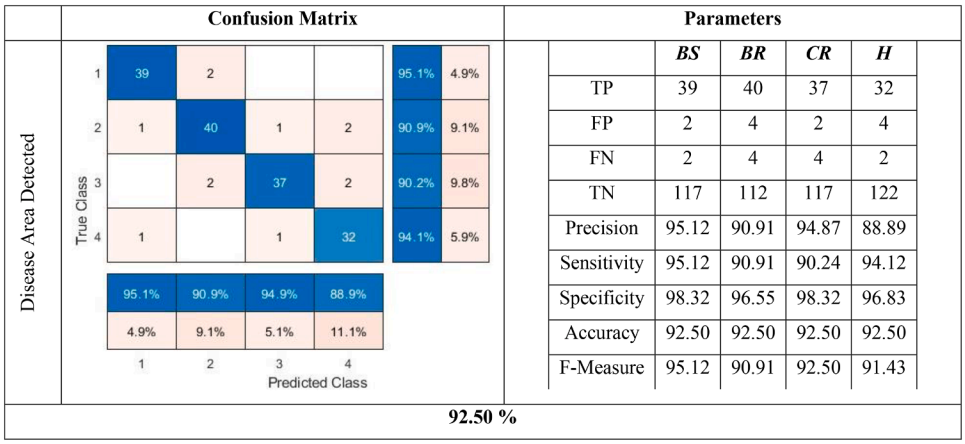


Fig. 8. Classification of apple leaf diseases using disease-specific regions: Confusion matrix (left) and evaluation criteria (right).

robust detection of true positive cases. Sensitivity for healthy leaves rose from 70.45 % to 84.21 %, showing enhanced recognition of non-diseased samples. Specificity followed a similar trend, with significant increases for both Black Rot and Cedar Rust, highlighting the model's improved ability to avoid false positives. The overall classification accuracy increased from 77.50 % to 83.13 %, and the F-measure showed marked improvement, particularly for Black Rot, which rose from 61.02 % to 82.19 %, reflecting better balance between precision and sensitivity.

These results clearly demonstrate that background removal enhances model performance by allowing more accurate and consistent disease classification, particularly for diseases previously prone to misclassification.

3.2.3. *Classification of apple leaf diseases using disease-specific regions*

The most substantial performance gains were achieved when the model focused specifically on disease-affected areas rather than analyzing the entire leaf. By isolating lesion regions following background removal, the model demonstrated exceptional precision across all classes: Black Spot (95.12 %), Black Rot (90.91 %), Cedar Rust (94.87 %), and Healthy leaves (88.89 %). These results indicate a significant reduction in false positives and enhanced disease detection accuracy. Sensitivity also improved markedly, especially for Black Spot and Black Rot, both reaching 95.12 % and 90.91 %, respectively, demonstrating the model's heightened ability to detect true disease instances. Specificity increased across all categories, with Black Spot and Cedar Rust both achieving 98.32 %, further validating the model's capability to distinguish diseased from healthy tissue. Classification accuracy reached 92.50 % for all classes, emphasizing consistent and robust performance. Similarly, the F-measure improved significantly, especially for Black Spot, Black Rot, and Cedar Rust, reflecting a well-balanced trade-off between precision and sensitivity.

These findings highlight the critical value of focusing on disease-specific regions for classification. This targeted approach, especially when combined with background removal, enables the model to more accurately and reliably differentiate between healthy and diseased tissue. Overall, disease area detection proved to be a highly effective strategy for enhancing automated plant disease diagnostics.

3.2.4. *Comparative analysis, key contributions, and limitations*

Although leaf disease symptoms often exhibit visual similarities, the classification results presented in this study demonstrate competitive accuracy when compared to prior research. The observed differences in classification performance relative to other studies can be attributed to several factors, including the accuracy achieved, the diversity and number of extracted features, the choice of classification algorithm, and the size and quality of the dataset used for training.

A critical consideration in plant disease classification is the degree of visual similarity among different diseases, both within the same species and across different plant types. For example, early-stage symptoms of apple scab may resemble signs of fungal infections like *Alternaria* or *bacterial blight* in other crops, making accurate diagnosis challenging. Many leaf diseases, regardless of their biological cause, produce necrotic lesions, chlorotic rings, or irregular discolorations that are visually alike [42]. These similarities can lead to confusion not only for human experts but also for machine learning models, especially when the features used

are limited or not sufficiently discriminative. Therefore, the robust performance of our model, even in the presence of such visual ambiguities, highlights the effectiveness of combining texture, color, and shape features in a complementary manner.

Moreover, the general visual patterns observed in infected leaves, such as spot shape irregularity, edge sharpness, and color transitions, are not unique to apple trees and are widely present in various plant species affected by foliar diseases. In grapevine, for example, downy mildew (*Plasmopara viticola*) causes yellowish oily spots on the upper surface and white fungal growth on the underside, which can resemble early symptoms of bacterial blight in other crops. In pomegranate, bacterial leaf spot leads to small dark necrotic lesions with a halo, which can easily be confused with *Alternaria* infections in tomato or *Cercospora* leaf spots in sugar beet. Tomato plants suffering from early blight (*Alternaria solani*) show concentric ring patterns within brown lesions, a symptom visually similar to target spot in pepper or even brown leaf spot in rice [38,43].

In potato crops, late blight (*Phytophthora infestans*) causes large, irregular water-soaked spots with chlorotic borders that mimic bacterial soft rot lesions. Strawberry leaves infected by leaf scorch (*Diplocarpon earlianum*) exhibit small, purplish spots that merge into larger dead areas, which resemble Septoria leaf spot in tomato or angular leaf spot in cucumber. In maize, northern corn leaf blight (*Exserohilum turcicum*) produces long, cigar-shaped grayish lesions, similar in appearance to gray leaf spot or *Helminthosporium* infections in other cereal crops. Additionally, rust diseases in wheat and barley, leaf smut in sugarcane, and anthracnose in beans and mangoes all create visually comparable lesions, ranging from powdery pustules to sharply edged necrotic zones, across diverse crop types [44].

The presence of such overlapping symptoms across plant species and pathogens highlights the challenge of visual diagnosis and underscores the importance of feature-based machine learning methods. The proposed model, which relies on transferable and interpretable features such as texture uniformity, edge geometry, and color deviation, can be adapted and fine-tuned for these crops as well. Particularly in resource-limited farming systems, where expert evaluation or diagnostic equipment is unavailable, this method provides a scalable, explainable, and practical solution. Its modular design enables integration into low-cost agricultural monitoring tools for early disease detection in a wide variety of economically important crops.

A comparative performance of various artificial intelligence models applied to plant disease classification in recent years is summarized in Table 2. This table highlights the evolution of methodologies, dataset sizes, and achieved accuracies, providing a contextual framework to position the current study within the broader research landscape.

One of the persistent challenges in earlier studies has been the limited size of datasets available for training deep learning models, particularly convolutional neural networks (CNNs), which has constrained classifier performance [57,58]. Additionally, deep learning methods, often regarded as "black box" approaches, extract complex features within the network layers that are difficult to interpret, especially for researchers lacking an in-depth understanding of model internals [59,60]. In contrast, traditional machine learning methods, when paired with the careful extraction of interpretable features such as texture, color, and shape, offer a more transparent and often more practical alternative for plant disease classification [61,62]. This study emphasizes that even with relatively small datasets, machine learning models can achieve strong classification performance if the features are both accurate and relevant. The findings support previous work suggesting that precise feature engineering can compensate for limited training data [38,63].

A key contribution of this study lies in its use of diverse and complementary feature types. Rather than relying on a single group of features, as in many earlier machine learning-based studies [64], this work integrates texture, color (from multiple color spaces), and shape features. This multi-feature approach, combined with color image processing and ACO for feature selection, led to robust and accurate identification of diseased regions on apple leaves. The analysis revealed that all three feature groups (texture, color, and shape) contribute significantly and almost equally to classification performance. Omitting any of these groups resulted in a noticeable decline in model accuracy, underscoring the importance of a holistic feature selection strategy.

Despite these strengths, the study also has limitations. Notably, the system struggles to detect diseases in their early stages (e.g., within two to three days post-infection), when symptoms are either absent or visually indistinct. Furthermore, the limited spectral sensitivity of standard RGB cameras constrains the detection of subtle biochemical changes in infected leaves [65]. While hyperspectral imaging could address this limitation by providing richer spectral data, such systems remain prohibitively expensive and complex for widespread agricultural deployment [66]. The scope of disease categories examined in this study was also limited. Expanding the number and variety of disease classes in future research would enhance model generalizability and practical utility. Additionally, the modest dataset size restricted the application of data-intensive methods such as deep learning [67].

In summary, this study contributes to the field of plant disease diagnostics by demonstrating the effectiveness of integrating multiple interpretable feature groups with optimized feature selection. It provides a foundation for future work and offers a comparative benchmark

Table 2
Example of methods to diagnose apple tree diseases.

Number of Classes	Methods	Accuracy (%)	Reference
Four	Combining genetic algorithm (GA) and correlation based feature selection (CFS), and SVM classifier.	90	Chuanlei et al., [45]
Three	Color and texture feature extraction and SVM classification	91	Sivakamasundari, [46]
Five	Inception module and Rainbow concatenation	78.80	Jiang et al., [47]
Three	Region-of-interest-aware deep convolutional neural network (ROI-aware DCNN)	84.3	Yu et al., [48]
Four	An ensemble of the pre-trained neural networks DenseNet121, EfficientNetB7, and EfficientNet	90	Bansal et al., [49]
Two	Two-stage apple disease detection system based on Xception and Faster-RCNN	88	Khan et al., [50]
Four	DF-Tiny-YOLO, based on deep learning feature extraction	90.88	Di & Li, [51]
Five	Faster R-CNN	63.10	Gong & Zhang, [52]
Three	Improved YOLOv5s model. Feature pyramid network (BiFPN) and transformer, and convolutional block attention module (CBAM).	84.30	Li et al., [53]
Five	Improvement of the traditional Yolov5 detection network called A-Net	92.7	Liu & Li, [54]
Three	Removed the prediction head for tiny objects from TPH-YOLOV5	91.4	Li et al., [55]
Three	ResNet50 model	93.19	Li et al., [55]
Seven	Enhanced lightweight model, ELM-YOLOv8n	94	Wang et al., [56]
Four	ROI extraction with *Auto K*-means clustring, feature extraction (color, texture, and shape), ACO, and SVM. Using original images Using background-removed images Using disease-specific regions	 77.5 83.12 92.50	This study

for evaluating improvements in classification methods.

4. Advancing smart and sustainable agriculture through AI-Driven disease detection and IoT integration

The proposed method, combining ACO, AI-based classification, and advanced image processing, offers a powerful and scalable solution for modern agriculture, especially in the detection and management of apple leaf diseases such as Black Spot, Black Rot, and Cedar Rust. These diseases significantly threaten apple yield and fruit quality, making early and accurate detection essential for timely intervention and effective crop protection [68]. The image processing component of this approach enables the extraction of key visual features from leaf images, including color and texture, which are critical for distinguishing between disease types. The ACO algorithm then optimizes the selection of the most informative features, reducing dimensionality and computational load while improving classification accuracy. This synergistic integration not only enhances diagnostic precision but also facilitates real-time, automated disease detection, empowering growers to take prompt action and reduce crop losses.

The method's utility is further amplified when integrated with Internet of Things (IoT) technologies. IoT-enabled devices, such as smart cameras and environmental sensors, can continuously monitor crop conditions and capture high-resolution images along with contextual data (e.g., temperature, humidity, and leaf wetness) [69,70]. These data streams can be processed using AI models enhanced by ACO-driven feature selection, ensuring high-confidence disease classification under varying field conditions. Once a disease is identified, the system can issue automated alerts, enabling farmers to apply localized treatments promptly, thereby minimizing pesticide use and maximizing treatment effectiveness.

Beyond disease detection, the proposed IoT-integrated framework can serve as a cornerstone for precision agriculture. Real-time diagnostics combined with environmental data allow for dynamic farm management. For example, upon detecting conditions favorable for Black Spot proliferation, such as elevated humidity and temperature, the system could initiate alerts for preventive fungicide applications. This closed-loop, data-informed approach enhances operational efficiency, ensures timely responses, and optimizes resource use, enhancing agricultural efficiency [71,72]. The integration of ACO-based AI diagnostics with IoT infrastructure provides a holistic and intelligent agricultural management system. This approach not only strengthens disease surveillance and response but also promotes resource-efficient practices, contributing to greater food security, environmental sustainability, and economic resilience in the agricultural sector.

5. Conclusion

This study presents a robust and interpretable framework for plant disease classification by integrating image processing, ACO-based feature selection, and machine learning classifiers. By extracting and selecting the most informative features, spanning texture, color, and shape, the model effectively distinguished visually similar apple leaf diseases, including Black Spot, Black Rot, and Cedar Rust. Preprocessing techniques, particularly background removal and disease area segmentation, significantly enhanced the model's precision and sensitivity by focusing classification on infected regions and reducing the impact of noise and visual ambiguity.

The results highlight the critical role of combining expert-driven feature engineering with intelligent selection algorithms in developing practical, high-accuracy diagnostic systems, especially when large labeled datasets for deep learning are unavailable. With an overall accuracy of 92.5 %, the proposed method demonstrates its potential for real-world agricultural applications. When paired with IoT technologies, this approach can enable real-time disease monitoring and precision management, contributing to more sustainable farming practices and reduced crop losses. Future research may extend this framework to other crops and explore hybrid models that integrate the interpretability of machine learning with the scalability of deep learning.

Ethical statement

The authors ensure that all procedures were performed in compliance with relevant laws and institutional guidelines. No human or animal subjects are involved in this research.

CRediT authorship contribution statement

Xiaolong Li: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Feifan Huang:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Haotian Sun:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Jiayu He:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Seyed Mohamad Javidan:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Data curation, Conceptualization. **Yiannis Ampatzidis:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Zhao Zhang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Not applicable.

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